

Revision date: 23 Jan 2025 Version: 3 Print date: 12 Mar 2026

Safety Data Sheet according to Regulation (EC) No. 1907/2006 (REACH)

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name/designation:

LabClean™

Article No.:

15-4100

UFI:

N9WP-4T6Y-EJR8-2XUG

1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the substance/mixture:

The product is intended for research, analysis and scientific education.

Cleaning agent. Restricted to professional users.

Relevant identified uses:

Life cycle stage [LCS]

PW: Widespread use by professional workers

Sector of uses [SU]

SU 24: Scientific research and development

Product Categories [PC]

PC 35: Washing and cleaning products

1.3. Details of the supplier of the safety data sheet

Supplier (manufacturer/importer/only representative/downstream user/distributor):

Minerva Biolabs GmbH

Schkopauer Ring 13

12681 Berlin

Germany

Telephone: +49 30 2000437 0

E-mail: ehs@minerva-biolabs.com

Website: www.minerva-biolabs.com

1.4. Emergency telephone number

Vergiftungs-Informations-Zentrale Freiburg, 24h: +49 761 19240

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Skin corrosion/irritation (<i>Skin Corr. 1</i>)	H314: Causes severe skin burns and eye damage.	Calculation method.

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms:



GHS05

Corrosion

Signal word: Danger

Hazard statements for health hazards	
H314	Causes severe skin burns and eye damage.

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Supplemental hazard information: none

Precautionary statements Prevention

P264	Wash hands thoroughly after handling.
P280	Wear protective gloves/protective clothing and eye protection/face protection.

Precautionary statements Response

P301 + P330 + P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

2.3. Other hazards

No data available

SECTION 3: Composition/information on ingredients**3.2. Mixtures**

Hazardous ingredients / Hazardous impurities / Stabilisers:

Product identifiers	Substance name Classification according to Regulation (EC) No 1272/2008 [CLP]	Concentration
CAS No.: 7664-38-2 EC No.: 231-633-2 Index No.: 015-011-00-6 REACH No.: 01-2119485924-24	orthophosphoric acid Skin Corr. 1B (H314)  Danger Specific concentration limit (SCL) Skin Corr. 1B; H314: C ≥ 25% Skin Irrit. 2; H315: 10% ≤ C < 25% Eye Dam. 1; H318: C ≥ 25% Eye Irrit. 2; H319: 10% ≤ C < 25% Acute Toxicity Estimate ATE (oral) 1,530 mg/kg ATE (dermal) 2,740 mg/kg	< 60 weight-%
CAS No.: 68439-50-9	Fatty alcohol c12-c14, ethoxylated Aquatic Acute 1 (H400), Aquatic Chronic 3 (H412)  Warning M-factor (acute): 1 Acute Toxicity Estimate ATE (oral) > 2,000 mg/kg	< 10 weight-%

Full text of H- and EUH-phrases: see section 16.

SECTION 4: First aid measures*** 4.1. Description of first aid measures****General information:**

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Remove victim out of the danger area. Remove contaminated, saturated clothing immediately. Do not leave affected person unattended. If unconscious but breathing normally, place in recovery position and seek medical advice. Warning First aider: Pay attention to self-protection!

Following inhalation:

Provide fresh air. In case of respiratory tract irritation, consult a physician.

In case of skin contact:

After contact with skin, wash immediately with plenty of water and soap. Take off immediately all contaminated clothing. If skin irritation or rash occurs: Get medical advice/attention. Get immediate medical advice/attention.

After eye contact:

In case of contact with eyes flush immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart and consult an ophthalmologist.

Following ingestion:

If accidentally swallowed rinse the mouth with plenty of water (only if the person is conscious) and obtain immediate medical attention. Do NOT induce vomiting. Get immediate medical advice/attention. Rinse mouth immediately and drink 1 glass of water.

Self-protection of the first aider:

Use personal protection equipment.

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4.2. Most important symptoms and effects, both acute and delayed

Serious eye damage/eye irritation, Skin corrosion/irritation, Irritation to respiratory tract

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: Firefighting measures**5.1. Extinguishing media****Suitable extinguishing media:**

Water spray jet, Foam, Co-ordinate fire-fighting measures to the fire surroundings.

Unsuitable extinguishing media:

Full water jet

5.2. Special hazards arising from the substance or mixture

The product itself does not burn.

Hazardous combustion products:In case of fire: Gases/vapours, toxic. Nitrogen oxides (NO_x), Phosphorus oxides, Carbon dioxide (CO₂)**5.3. Advice for firefighters**

Fight fire with normal precautions from a reasonable distance. Wear a self-contained breathing apparatus and chemical protective clothing. Do not breathe gas/fumes/vapour/spray.

5.4. Additional information

Collect contaminated fire extinguishing water separately. Do not allow entering drains or surface water.

SECTION 6: Accidental release measures**6.1. Personal precautions, protective equipment and emergency procedures****6.1.1. For non-emergency personnel****Personal precautions:**

Remove persons to safety. Use personal protective equipment as required. Avoid contact with skin, eyes and clothes.

Protective equipment:

Wear protective gloves/protective clothing and eye protection/face protection.

6.1.2. For emergency responders**Personal protection equipment:**

Personal protection equipment: see section 8

6.2. Environmental precautions

Do not allow to enter into surface water or drains. Prevent spread over a wide area (e.g. by containment or oil barriers).

6.3. Methods and material for containment and cleaning up**For containment:**

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents). Treat the recovered material as prescribed in the section on waste disposal.

For cleaning up:

Water (with cleaning agent)

6.4. Reference to other sections

Safe handling: see section 7, Personal protection equipment: see section 8, Disposal: see section 13

6.5. Additional information

Use appropriate container to avoid environmental contamination.

SECTION 7: Handling and storage*** 7.1. Precautions for safe handling****Protective measures****Advices on safe handling:**

Wear personal protection equipment (refer to section 8). Wash hands before breaks and after work. Keep away from: Food and feedingstuffs. If local exhaust ventilation is not possible or not sufficient, the entire working area should be ventilated by technical means.

Fire prevent measures:

Usual measures for fire prevention.

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Advices on general occupational hygiene

Avoid contact with skin, eyes and clothes. When using do not eat, drink or smoke. Avoid contact with eyes and skin.

7.2. Conditions for safe storage, including any incompatibilities**Technical measures and storage conditions:**

Keep container tightly closed in a cool, well-ventilated place.

Packaging materials:

Keep/Store only in original container.

Requirements for storage rooms and vessels:

Keep container tightly closed in a cool, well-ventilated place. Recommended storage temperature: at room temperature.

Keep away from: Food and feedingstuffs

Hints on storage assembly:

Do not store together with: alkali, Food and feedingstuffs

Storage class (TRGS 510, Germany): 8B – Non-combustible corrosive substances

7.3. Specific end use(s)

No data available

SECTION 8: Exposure controls/personal protection*** 8.1. Control parameters****8.1.1. Occupational exposure limit values**

Limit value type (country of origin)	Substance name	① Long-term occupational exposure limit value ② Short-term occupational exposure limit value ③ Instantaneous value ④ Monitoring and observation processes ⑤ Remark
TRGS 900 (DE)	orthophosphoric acid CAS No.: 7664-38-2 EC No.: 231-633-2	① 2 mg/m ³ ② 4 mg/m ³ ⑤ (einatembare Fraktion) DFG, EU, AGS, Y
IOELV (EU)	orthophosphoric acid CAS No.: 7664-38-2 EC No.: 231-633-2	① 1 mg/m ³ ② 2 mg/m ³

8.1.2. Biological limit values

No data available

8.1.3. DNEL-/PNEC-values

Substance name	DNEL value	① DNEL type ② Exposure route
orthophosphoric acid CAS No.: 7664-38-2 EC No.: 231-633-2	1 mg/m ³	① DNEL worker ② Long-term – inhalation, local effects
orthophosphoric acid CAS No.: 7664-38-2 EC No.: 231-633-2	2 mg/m ³	① DNEL worker ② Acute - inhalation, local effects

8.2. Exposure controls**8.2.1. Appropriate engineering controls**

If local exhaust ventilation is not possible or not sufficient, the entire working area should be ventilated by technical means.

8.2.2. Personal protection equipment**Eye/face protection:**

Eye glasses with side protection (EN 166).

Skin protection:

Tested protective gloves must be worn (EN ISO 374). Take recovery periods for skin regeneration. Breakthrough times and swelling properties of the material must be taken into consideration. Suitable material: NBR (Nitrile rubber), CR

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(polychloroprene, chloroprene rubber), FKM (fluoro rubber). Thickness of the glove material: > 0,35 mm. Breakthrough time: >480 min. In the case of wanting to use the gloves again, clean them before taking off and air them well.

Respiratory protection:

Usually no personal respirative protection necessary.

Other protection measures:

Wash hands before breaks and after work.

8.2.3. Environmental exposure controls

No data available

SECTION 9: Physical and chemical properties**9.1. Information on basic physical and chemical properties****Appearance**

Physical state: Liquid

Colour: light yellow

Odour: not determined

flammability: No

Safety relevant basis data

Parameter	Value	at °C	① Method ② Remark
pH	0.5	20 °C	① DGF H-III 1
Melting point	-20 °C		
Freezing point	No data available		
Initial boiling point and boiling range	100 °C		
Flash point	not applicable		
Evaporation rate	No data available		
Auto-ignition temperature	not applicable		
Upper/lower flammability or explosive limits	No data available		
Vapour pressure	No data available		
Vapour density	No data available		
Density	1.36 g/cm³	20 °C	① DIN 12791
Bulk density	not applicable		
Water solubility	completely miscible		
Dynamic viscosity	No data available		
Kinematic viscosity	No data available		

9.2. Other information

No data available

SECTION 10: Stability and reactivity**10.1. Reactivity**

No specific test data related to reactivity available for this product or its ingredients. The product itself does not burn.

10.2. Chemical stability

The product is chemically stable under recommended conditions of storage, use and temperature.

10.3. Possibility of hazardous reactions

No known hazardous reactions.

10.4. Conditions to avoid

Further information on proper storage: see section 7. Thermal decomposition can lead to the escape of irritating gases and vapours.

10.5. Incompatible materials

Alkali (lye), concentrated, Alkali metals

10.6. Hazardous decomposition products

No known hazardous decomposition products.

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SECTION 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

orthophosphoric acid	CAS No.: 7664-38-2	EC No.: 231-633-2
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LD₅₀ oral: 1,530mg/kg (Rat)

LD₅₀ dermal: 2,740mg/kg (Rabbit)
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Fatty alcohol c12-c14, ethoxylated	CAS No.: 68439-50-9
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LD₅₀ oral: >2,000mg/kg (Rat)
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Acute oral toxicity:

Based on available data, the classification criteria are not met.

Acute dermal toxicity:

Based on available data, the classification criteria are not met.

Acute inhalation toxicity:

Based on available data, the classification criteria are not met.

Skin corrosion/irritation:

Causes severe skin burns and eye damage.

Serious eye damage/irritation:

Causes serious eye damage.

Respiratory or skin sensitisation:

Based on available data, the classification criteria are not met.

Germ cell mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity:

Based on available data, the classification criteria are not met.

STOT-single exposure:

Based on available data, the classification criteria are not met.

STOT-repeated exposure:

Based on available data, the classification criteria are not met.

Aspiration hazard:

Based on available data, the classification criteria are not met.

Additional information:

No data available

11.2. Information on other hazards

No data available

SECTION 12: Ecological information

12.1. Toxicity

orthophosphoric acid	CAS No.: 7664-38-2	EC No.: 231-633-2
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LC₅₀: 138mg/L 4d (fish, <i>Gambusia affinis</i> (Belebttschlamm))

EC₅₀: >100mg/L 2d (crustaceans)

EC₅₀: >100mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i> (previous name: <i>Scenedesmus subspicatus</i>))
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EC₅₀: >100mg/L 2d (crustaceans, <i>Daphnia magna</i>)
--

NOEC: 100mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i> (previous name: <i>Scenedesmus subspicatus</i>))
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NOEC: 56mg/L 2d (crustaceans, <i>Daphnia magna</i>)

ErC₅₀: >100mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i>)

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Fatty alcohol c12-c14, ethoxylated CAS No.: 68439-50-9LC₅₀: 0.423mg/L 4d (fish, Pimephales promelas) OECD Guideline 203 (Fish, Acute Toxicity Test)EC₅₀: 0.044mg/L 3d (Algae/water plant, Raphidocelis subcapitata (previous names: Pseudokirchneriella subcapitata, Selenastrum capricornutum)) OECD Guideline 201 (Alga, Growth Inhibition Test)EC₅₀: 0.125mg/L 2d (crustaceans, Daphnia magna) OECD Guideline 202 (Daphnia sp. Acute Immobilisation Test)

NOEC: <1mg/L (Algae/water plant)

NOEC: 0.037mg/L 3d (Algae/water plant, Raphidocelis subcapitata (previous names: Pseudokirchneriella subcapitata, Selenastrum capricornutum)) OECD Guideline 201 (Alga, Growth Inhibition Test)

NOEC: 0.304mg/L 4d (fish, Pimephales promelas) OECD Guideline 203 (Fish, Acute Toxicity Test)

NOEC: 0.045mg/L 2d (crustaceans, Daphnia magna) OECD Guideline 202 (Daphnia sp. Acute Immobilisation Test)

12.2. Persistence and degradability

No data available

12.3. Bioaccumulative potential**Fatty alcohol c12-c14, ethoxylated** CAS No.: 68439-50-9Log K_{OW}: 7**12.4. Mobility in soil**

No data available

12.5. Results of PBT and vPvB assessment**orthophosphoric acid** CAS No.: 7664-38-2 EC No.: 231-633-2

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

Fatty alcohol c12-c14, ethoxylated CAS No.: 68439-50-9

Results of PBT and vPvB assessment: This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

12.6. Endocrine disrupting properties

No data available

12.7. Other adverse effects

No data available

SECTION 13: Disposal considerations**13.1. Waste treatment methods**

Dispose of waste according to applicable legislation.

13.1.1. Product/Packaging disposal**Waste codes/waste designations according to EWC/AVV****Waste code product**

20 01 29 * Detergents containing hazardous substances

*: Evidence for disposal must be provided.

Waste treatment options**Appropriate disposal / Product:**

Consult the appropriate local waste disposal expert about waste disposal.

Appropriate disposal / Package:

Completely emptied packages can be recycled.

SECTION 14: Transport information

Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.1. UN number or ID number			
UN 1805	UN 1805	UN 1805	UN 1805
14.2. UN proper shipping name			
PHOSPHORIC ACID, SOLUTION	PHOSPHORIC ACID, SOLUTION	PHOSPHORIC ACID, SOLUTION	PHOSPHORIC ACID, SOLUTION
14.3. Transport hazard class(es)			
 8	 8	 8	 8

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Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.4. Packing group			
III	III	III	III
14.5. Environmental hazards			
No	No	No	No
14.6. Special precautions for user			
Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 Hazard identification number (Kemler No.): 80 Classification code: C1 Tunnel restriction code: (E)	Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 Classification code: C1	Special Provisions: 223 Limited quantity (LQ): 5 L Excepted Quantities (EQ): E1 EmS-No.: F-A, S-B	Special Provisions: A3 Limited quantity (LQ): Y841 Excepted Quantities (EQ): E1

14.7. Maritime transport in bulk according to IMO instruments

No data available

SECTION 15: Regulatory information**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****15.1.1. EU legislation****Other regulations (EU):**

1907/2006 REACH, 1272/2008 CLP GHS, 98/24/EC

15.1.2. National regulations **[DE] National regulations****Restrictions of occupation**

22 JArbSchG., 4 MuSchRiV., 5 MuSchRiV., Betriebssicherheitsverordnung (BetrSichV)

Water hazard class**WGK:**

2 - obviously hazardous to water

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Other regulations, restrictions and prohibition regulations

Hazardous Substances Ordinance (GefStoffV)

15.2. Chemical Safety Assessment

For this substance a chemical safety assessment has not been carried out.

SECTION 16: Other information**16.1. Indication of changes**

4.1.	Description of first aid measures
7.1.	Precautions for safe handling
8.1.	Control parameters

16.2. Abbreviations and acronyms

For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemical safety assessment, chapter R.20 (Table of terms and abbreviations).

16.3. Key literature references and sources for data

TRGS 510, TRGS 525, TRGS 900, Safety data sheets for the ingredients, 1907/2006 REACH, 1272/2008 CLP GHS

Substance name	Type	source of supply
orthophosphoric acid CAS No.: 7664-38-2 EC No.: 231-633-2	EC ₅₀ ; NOEC	Source: European Chemicals Agency, http://echa.europa.eu/
Fatty alcohol c12-c14, ethoxylated CAS No.: 68439-50-9	LC ₅₀ ; EC ₅₀ ; NOEC	Source: European Chemicals Agency, http://echa.europa.eu/

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16.4. Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Skin corrosion/irritation (<i>Skin Corr. 1</i>)	H314: Causes severe skin burns and eye damage.	Calculation method.

16.5. List of relevant hazard statements and/or precautionary statements from sections 2 to 15

Hazard statements	
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H400	Very toxic to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

16.6. Training advice

No data available

16.7. Additional information

This data sheet was prepared in accordance with EU Regulation (EC) No. 1907/2006 (REACH) and, where applicable, US Regulation 29 CFR 1910.1200, Appendix D.

The information is based on the level of knowledge and experience on the date of issue, it do not have the importance of assured properties. The information may not be changed or transferred to other products. Reproduction in the unchanged state is permitted.

* Data changed compared with the previous version.